Aditya Sunke

Blacksburg, VA | Email | LinkedIn | GitHub

EDUCATION

Virginia Tech, Blacksburg, VA Expected Graduation: May 2027
Major: B.S. Computer Science (C.S.) | **Minor:** Quantum Information Science and Engineering (Q.I.S.E.)
GPA: 3.81/4.0 | **Dean's List:** Fall 2023, Spring, 2024, Fall 2024, Spring 2025
Relevant Coursework: Data Structures and Algorithms, Intermediate Software Design, Computer Organization, Computer Systems, Problem Solving in CS, Applied Combinatorics, Graph Theory, Quantum Software, Discrete Mathematics

WORK EXPERIENCE

Co-founder and Quantum Lead **July 2025 – Present**
CarbonQapture Blacksburg, VA

- Co-founded and led a 7-member interdisciplinary team focused on applying quantum computing and AI to carbon capture technologies, securing over $300 in university funding each semester to support R&D efforts.
- Designing and implementing quantum chemistry simulations using VQE on Qiskit and Cirq to simulate Carbon Dioxide with Metal Organic Frameworks enabling near-term quantum application testing.

Research Assistant **Mar 2025 – July 2025**
National Science Foundation Blacksburg, VA

- Advanced cryptographic research under a $247K NSF-funded grant by analyzing multivariate Goppa code properties and prototyping their integration into the McEliece cryptosystem, strengthening resilience against quantum attacks.
- Integrated multivariate Goppa codes into the McEliece cryptosystem, reducing key size and improving decoding efficiency contributing to NIST standardization efforts.

AI Developer **Jan 2025 – May 2025**
Virginia Tech College of Engineering Blacksburg, VA

- Collaborated within an interdisciplinary team to design an AI-assisted navigation system enabling visually impaired individuals to explore tourist and museum spaces independently.
- Developed and deployed a prototype wearable Raspberry Pi–based necklace integrating Microsoft Azure Computer Vision and a custom-trained Google Gemini model on museum artifacts for real-time environmental understanding.

PROJECTS

Soteria **– Full-stack Women's Safety App** **Jan 2026**

- Developed a full-stack React Native (Expo + TypeScript) mobile app with modular service-layer architecture for authentication, location tracking, session management, and safety-circle coordination.
- Integrated Firebase, mapping and routing APIs, and EmailJS for GPS tracking, SOS alerts, and automated notification.
- Implemented an AI-powered chatbot using Google Gemini API to help users use features of app through natural language.

SilentVoice **– Real-Time ASL Word and Finger-spelling Translation** **Dec 2025 – Present**

- Developing a computer vision and machine learning system to translate sign language gestures into natural language sentences, framing translation as a sequence modeling problem under visual uncertainty.
- Developed a CNN-based finger-spelling recognition model to classify hand gesture images into alphabet characters, enabling real-time translation of sign language finger spelling into text.

Qure AI **– Quantum/AI Approach to Drug Discovery for Meningitis** **Sept 2025**

- Developed an SVM-based drug classification model achieving 90%+ prediction accuracy across four meningitis-causing bacterial datasets and applied Variational Quantum Eigensolver (VQE) algorithm to simulate molecular ground-state energies, validating stability of machine-learning-generated compounds.
- Generated 22 novel drug candidates with ground-state energies 1.98× lower than existing drugs, indicating significantly higher molecular stability.

TECHNICAL SKILLS

Languages & Frameworks | Python, Java, C/C++, JavaScript, TypeScript, SQL, RISC-V Assembly, NumPy, SciPy, Pandas, PyTorch, scikit-learn, OpenCV, Matplotlib, PySCF, React Native, Expo, Firebase, Leaflet
Developer Tools | Git, Docker, Linux, Jupyter, Hugging Face, npm, Node.js, Firebase Console, Google Cloud Console, Google Places API, OSRM Routing, OpenStreetMap, Google Gemini API, EmailJS
Skills | Machine learning, computer vision, quantum computing, numerical optimization, algorithm design, mobile app development, cloud applications, software architecture, modular design